ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

PRINCIPLES OF COMPUTER PROGRAMMING

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

NC(V) Information Technology and Computer Science, Level 3. Section A: objective questions (30) · Section B: programming concepts, data types and expressions, selection and loops, arrays, procedures and functions, debugging (70).

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INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Code in this paper is written in Python. You may write your answers in Python, C#, Java or pseudocode – state which language you use. Syntax will not be penalised if the logic is correct.
5. Write neatly and legibly. Indent your code clearly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 A program that translates and executes source code one line at a time is a/an …

A interpreter.
B compiler.
C assembler.
D linker.

1.1.2 The result of the Python expression 7 // 2 is …

A 3.5.
B 4.
C 1.
D 3.

1.1.3 The result of the expression 2 + 3 * 4 ** 2 is …

A 80.
B 50.
C 400.
D 14.

1.1.4 The first element of an array (list) named marks is accessed with …

A marks[1].
B marks(first).
C marks[-0.5].
D marks[0].

1.1.5 Which loop is the best choice when you do not know in advance how many times it must repeat?

A A FOR loop over a fixed range
B A conditional (WHILE) loop
C A nested IF
D A function call

1.1.6 A variable that is declared inside a function and can only be used there is a … variable.

A global
B constant
C local
D public

1.1.7 The value of not (5 > 3 and 2 == 2) is …

A False.
B True.
C 5.
D an error.

1.1.8 Converting the string "25" into the integer 25 is called …

A concatenation.
B type casting (conversion).
C iteration.
D encapsulation.

1.1.9 Joining two strings together, for example "Cape" + "Town", is …

A slicing.
B indexing.
C rounding.
D concatenation.

1.1.10 Dividing a number by zero while the program is running causes a …

A runtime error.
B syntax error.
C logic error only.
D compilation warning only.

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 A function always returns a value, while a procedure does not.

1.2.2 An infinite loop is a loop whose condition never becomes false.

1.2.3 Arrays (lists) can store many values under one name.

1.2.4 In Python, = is used to compare two values for equality.

1.2.5 Comments are ignored when the program is executed.

1.2.6 len("NCV L3") returns 6.

1.2.7 A nested loop is a loop inside another loop.

1.2.8 Parameters allow values to be passed into a function.

1.2.9 Meaningful variable names make a program harder to maintain.

1.2.10 A boundary value is a value at the edge of a valid range.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	Human-readable instructions written by a programmer	A	IDE
1.3.2	Instructions in binary that the CPU can execute directly	B	source code
1.3.3	A value passed to a function when it is called	C	machine code
1.3.4	Software for writing, running and debugging code in one place	D	modularisation
1.3.5	Checking that input is sensible before it is used	E	stepping
1.3.6	Running the program one line at a time to find errors	F	breakpoint
1.3.7	Stopping point set in the code during debugging	G	validation
1.3.8	A value that cannot change while the program runs	H	constant
1.3.9	Position number of an element in an array	I	argument
1.3.10	Breaking a large program into smaller parts	J	index

(10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2: PROGRAMMING CONCEPTS AND EXPRESSIONS

2.1 Explain the difference between a compiler and an interpreter. Give ONE advantage of each. (4)

2.2 Give the output of each of the following Python statements: (a) print(17 % 5) (b) print("ab" * 3) (c) print(round(3.678, 1)) (d) print(len("Mthatha")) (4)

2.3 Write a single Boolean expression that is True when a variable age is between 18 and 35 (inclusive) AND a variable hasID is True. (3)

2.4 Explain THREE rules or conventions for naming variables. (3)

2.5 Name TWO programming paradigms (styles) and briefly describe ONE of them. (3)

[17]

QUESTION 3: SELECTION AND LOOPS

3.1 A college awards results as follows: 80–100 Distinction, 70–79 Merit, 50–69 Pass, 0–49 Fail. Write code that asks for a mark, checks that it is between 0 and 100 (display "Invalid" if not), and displays the result. (8)

3.2 Study the code below and answer the questions.

```
total = 0
n = 1
while n <= 10:
    if n % 3 == 0:
        total = total + n
    n = n + 1
print(total)
```

3.2.1 What is the output of the code? Show how you got the answer. (3)

3.2.2 What would happen if the line n = n + 1 were removed? (2)

3.2.3 Rewrite the code using a FOR loop. (3)

3.3 Write code that keeps asking the user for a PIN until the correct PIN 4321 is entered, or until three attempts have been made. Display "Access granted" or "Card blocked". (6)

[22]

QUESTION 4: ARRAYS (LISTS)

4.1 The daily temperatures (°C) for a week in Kimberley are stored in a list:

```
temps = [31, 34, 29, 36, 33, 28, 30]
```

4.1.1 What is the value of temps[3]? (1)

4.1.2 Write code using a loop (not built-in max/min/sum functions) to find and display the highest temperature and the average temperature. (7)

4.1.3 Write code to count and display how many days were hotter than 30 °C. (3)

4.2 Explain the difference between a one-dimensional and a two-dimensional array. Give an example of data suited to a two-dimensional array. (3)

[14]

QUESTION 5: FUNCTIONS AND DEBUGGING

5.1 Write a function called calcVAT that takes a price excluding VAT as a parameter and returns the price including 15% VAT. Then show how the function is called for a price of R200 and the answer displayed. (6)

5.2 Give THREE advantages of using functions (sub-programs). (3)

5.3 The code below should calculate the average of three marks, but it contains errors. Identify THREE errors and correct them.

```
1  mark1 = input("Mark 1: ")
2  mark2 = int(input("Mark 2: "))
3  mark3 = int(input("Mark 3: ")
4  average = mark1 + mark2 + mark3 / 3
5  print("Average: " + average)
```

(6)

5.4 Classify each error in QUESTION 5.3 as a syntax, runtime or logic error. (2)

[17]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER